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Probabilistic Sub-Daily Streamflow Forecasting in Data-Scarce Catchments Through a Zero-Shot Inductive Transfer Learning Approach

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ABSTRACT

Forecasting sub-daily streamflow in regions with limited discharge data, such as newly gauged or sparsely monitored catchments, poses challenges to Deep Learning (DL) models, which require large training datasets. These scenarios, known as Prediction in Newly Gauged Basins (PNB) or Prediction with Limited Data (PLD), require models that remain accurate even when local training data are limited. This study uses the Temporal Fusion Transformer (TFT), a DL model that captures complex dependencies and generates probabilistic forecasts, supporting hydrological decision-making under uncertainty. A single TFT model trained on 15 diverse gauged catchments was evaluated for a zero-shot generalization on 10 unseen catchments using hydrological, meteorological, and physical attributes (catchments located across the eastern and south-eastern United States), simulating real-world PNB/PLD conditions. Results showed that the TFT model outperformed the National Water Model in continuous hydrograph simulation and flood forecasting. It achieved higher accuracy (average NSE = 0.655 vs. 0.58 for NWMv3.0), with over 91% of observed values falling within its 95% prediction intervals. The model identifies dominant hydrologic drivers, predicts short-term floods with high accuracy across lead times, and removes the need for catchment-specific model training and tuning, thereby simplifying model deployment and improving scalability for operational hydrology.

1 | Introduction

Reliable streamflow and flood forecasting has become a central concern in hydrology due to increasing climate variability, extreme precipitation events, and the growing societal dependence on water-related infrastructure. Accurate forecasts are essential for flood risk management, hydropower scheduling, reservoir operations, and early warning systems that protect lives and property (Tellman et al. 2021; Arsenault and Côté 2019). However, changing climate patterns and intensifying hydrologic extremes are challenging the limits of traditional forecasting systems (Beven 2019; Razavi and Coulibaly 2013; Parajka et al. 2013). Moreover, physically based models like SAC-SMA typically demand substantial local calibration and high-resolution input data, making them costly and difficult to generalize to regions (Newman et al. 2017). To overcome the limitation of the traditional models, the hydrology community has increasingly adopted data-driven methods, particularly machine learning (ML) and deep learning (DL), which excel at modeling complex, nonlinear hydrologic processes without the need for explicit physical formulations. While these approaches have shown impressive predictive capabilities (Nevo et al. 2022; Sadeghi Tabas et al. 2023; Zhang et al. 2024), they typically require extensive historical data and involve computationally demanding training and tuning procedures. The extensive data requirement and high cost of data collection coupled with the substantial computational resources and time needed to calibrate these models limits their applicability in many real-world

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scenarios such as urgent operation needs or where streamflow observations are available in limited quantity, for example, in newly gauged or partially monitored basins. These situations fall into a growing category of hydrological challenges where available streamflow data is insufficient for training or fine-tuning data-driven models, yet still valuable for real-time forecasting. This intermediate condition, between fully gauged and ungauged, has been recognized as Prediction in Newly Gauged Basins (PNB) or Prediction with Limited Data (PLD) (Yin et al. 2023). Given the increasing demand for accurate and efficient hydrological forecasts in regions with emerging or sparse monitoring infrastructure, developing robust and generalizable models that can transfer knowledge from well-gauged to PNB or PLD basins is critical.

ML/DL methods have sparked a paradigm shift in how the hydrology community addresses the problem of Prediction in Ungauged Basins (PUB). These data-driven models can learn complex nonlinear relationships from historical data without needing explicit physical parameterization. Ghobadi et al. (2024) explored the long-term streamflow forecasting in data-scarce basins by combining satellite-derived inputs, the Informer Architecture (IA), and concurrent fine-tuning transfer learning demonstrating strong performance across hydrologically diverse regions without relying on dense in situ observations. Others, such as Frame et al. (2022), have explored ensemble and multitask learning strategies to improve generalization to new hydrological regimes, and there are other recent research that emphasize the importance and hurdles of the problems associated with PUB (Blöschl et al. 2019; Jahanshahi et al. 2021; Yoon et al. 2022; Arsenault et al. 2023). Recurrent Neural Networks (RNNs) have gained wide use in hydrology due to their ability to model nonlinear temporal dependencies and outperform process-based models in many streamflow and flood forecasting tasks (Heidari et al. 2025; Kumar et al. 2023; Sadeghi Tabas and Samadi 2022). Among RNNs, Long Short-Term Memory (LSTM) networks have shown promise in modeling streamflow dynamics, even outperforming process-based models in many cases (Kratzert et al. 2019). Studies conducted by Zhou et al. (2023) and Yang et al. (2023) have demonstrated that LSTM and graph-based DL models can capture spatiotemporal dependencies across river networks, offering strong generalization in ungauged basins. Beyond streamflow forecasting, DL is increasingly being applied to sub-daily and flood hydrograph forecasting in ungauged basins (PUB), where the absence of ground-based discharge data presents a formidable challenge. Nearing et al. (2024) showed that AI-based models can reliably predict extreme floods in ungauged basins, offering earlier and more accurate warnings than traditional methods. Sun et al. (2022) introduced a graph convolutional network (GCN) approach for flood peak prediction, leveraging the spatial structure of river networks to transfer learned behaviors from gauged to ungauged basins.

RNNs typically produce deterministic point forecasts, offering little insight into predictive uncertainty that is essential for flood risk management, reservoir operations, and other high-stakes decision-making scenarios (Gal and Ghahramani 2016). In addition, RNNs lack inherent interpretability mechanisms, making it difficult to understand which input variables (e.g., meteorological drivers or static catchment characteristics) are influencing predictions, a drawback that can reduce model transparency and limit their usefulness in real-world hydrologic applications (Waqas and Humphries 2024). To address these limitations, recent advances in sequence modeling have introduced attention mechanisms, most notably in the Transformer architecture by Vaswani et al. (2017). Attention allows models to dynamically weigh the importance of different input elements, thereby improving both performance and interpretability across long temporal sequences. For example, Xu et al. (2023) applied a Transformer-based model to predict flood discharge at 6- and 12-h lead times in data-sparse basins, using a range of meteorological and hydrological inputs. Similarly, Yin et al. (2023) explored the use of pre-trained DL models, including Transformer variants, for runoff prediction under limited data conditions. These studies reflect the growing interest in adapting Transformer architectures for hydrological forecasting, though they continue to rely on the availability of discharge data and often require model adjustments tailored to specific application scenarios.

Given the importance of capturing temporal patterns in hydrology, Transformers are sometimes complemented by recurrent layers such as LSTMs. This hybrid modeling strategy is evident in recent studies that integrate sequence-aware components or restructure inputs to better handle time-dependent features (Shi et al. 2024; Castangia et al. 2023). Temporal Fusion Transformer (TFT), a recent DL architecture developed for interpretable, multivariate time-series forecasting (Lim et al. 2021), builds on the Transformer paradigm and further enhances it for structured time-series problems. TFT combines the sequential modeling strength of LSTM layers with the flexibility of attention mechanisms, enabling it to effectively learn both temporal dynamics and feature importance. However, unlike a simple hybrid of LSTM and Transformer modules, TFT integrates several additional components tailored for real-world forecasting challenges. These include variable selection networks that dynamically identify the most relevant input features, static covariate encoders to incorporate time-invariant catchment characteristics, and gated residual networks (GRNs) that enhance nonlinear transformations while controlling information flow. The architecture also features a temporal fusion decoder that merges short- and long-range information across time steps in an interpretable manner. It incorporates multi-head attention, temporal gating, and probabilistic output layers, generating forecasts as quantiles rather than single-point estimates. This enables TFT to produce predictive distributions that directly quantify uncertainty, an essential requirement for risk-aware hydrologic forecasting, particularly in data-scarce or operational contexts. Although the TFT has demonstrated strong potential in various domains, its application in hydrology remains limited with only a handful of recent studies exploring its use for streamflow and flood forecasting (Koya and Roy 2024; Burrichter et al. 2024; Huang et al. 2024; Sadeghi Tabas et al. 2025). Unlike conventional deterministic models, TFT generates probabilistic forecasts, typically in the form of quantiles, allowing for full predictive distributions that directly quantify uncertainty. This is particularly valuable in PNB/PLD, where uncertainty is high and actionable, interpretable predictions are essential.

Building on these foundations, this study presents a novel approach to sub-daily streamflow forecasting for data-scarce

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catchments utilizing the TFT model. In this work, the forecasting challenge is framed as a zero-shot Inductive Transfer Learning (I-TL) with no target fine-tuning problem, in which a single TFT model is trained on a diverse set of gauged catchments using their meteorological, hydrological, and static catchment attributes. Kratzert et al. (2019) findings strongly suggest that there is sufficient information in static and dynamic inputs to support generalizable, data-driven streamflow prediction. This aligns with a broader movement in the hydrology community toward data-centric, model-agnostic learning approaches. The trained TFT model is then evaluated for its zero-shot cross-catchment generalization performance in completely unseen catchments without any fine-tuning or adaptation. While this does not represent the traditional PUB scenario (which assumes no observed discharge at all), the model's reliance on observed discharge and meteorological inputs at inference time reflects a practical setup often found in sparsely gauged catchments. This approach differs from conventional regionalization or fine-tuned transfer learning and serves as a true test of spatial generalization. To our knowledge, this is the first study to apply the TFT architecture to large-scale cross-catchment streamflow forecasting, leveraging its scalability and interpretability for hydrologic applications. The proposed method offers several key features: (i) Model innovation, through the deployment of TFT in a novel hydrological context; (ii) Generalization capability, as it achieves strong predictive performance in zero-shot settings, comparable to or exceeding LSTM-based baselines without calibration; (iii) Cost and deployment efficiency, by eliminating the need for per-catchment retraining, making it viable for large-scale and data-sparse operational contexts; and (iv) Explainability, as the TFT inherently highlights the most influential meteorological and static catchment features for each prediction, providing valuable transparency that is often lacking in DL applications to hydrology.

The structure of the paper is organized as follows: Section 1 offers state-of-the-art AI approaches applied within the context of data-scarce streamflow forecasting problems. Section 2 outlines the study area, data, and the algorithmic and training framework, including the benchmark model. Section 3 presents the hyperparameter tuning strategy and provides a comprehensive evaluation of the TFT's performance using both quantitative metrics, interpretability analysis, and uncertainty quantification. Finally, Section 4 concludes the findings, explores their implications, and proposes future directions for advancing short-term flood forecasting.

2 | Methodology

2.1 | Study Area and Data

In this study, 25 hydrologically diverse catchments distributed along the Atlantic and Gulf coasts are utilized, spanning from Maine to Louisiana. Of these, 15 catchments were designated for training and tuning the TFT model, while the remaining 10 catchments, kept completely unseen during training, were reserved for inference to evaluate the I-TL approach. For clarity, the catchments used for training are referred to as seen catchments and those used exclusively to assess the I-TL approach are labelled as unseen catchments throughout the remainder of this paper. The geographic distribution of these catchments is illustrated in Figure 1, where seen catchments are marked with red circles and unseen catchments with blue stars. This targeted selection is intended to facilitate real-time decision-making across a range of catchment types and to strengthen flood resilience in regions most at risk from future flood hazards.

The input data in this study come from three main sources:

1. The North American Land Data Assimilation System (NLDAS), providing hourly meteorological forcing data;
2. The USGS, supplying streamflow (discharge) data; and
3. The CAMELS dataset, which offers static physical characteristics of catchments.

To ensure a consistent benchmarking framework, the TFT model was forced using meteorological variables from the NLDAS reanalysis, aligning with the retrospective (reanalysis-based) forcing used by NWMv3.0, so that both models were evaluated under comparable atmospheric input conditions rather than real-time forecast forcings. An hourly dataset compiled by USGS and NLDAS is used. The dataset covers more than 500 catchments across the contiguous United States (Gauch et al. 2020). The CAMELS dataset, developed by the National Center for Atmospheric Research (NCAR), is used to describe each catchment's physical features. CAMELS contains data for 671 catchments, including attributes like soil type, long-term climate statistics, vegetation, topography, and geology. These features are crucial for simulating hydrological behavior and have made CAMELS a key resource for flood modeling studies (Addor et al. 2018; Newman et al. 2015; Kratzert et al. 2019). The TFT model was trained using data from the seen catchments spanning October 1997 to September 2014. To evaluate the model's generalization capabilities, the subsequent period, October 2014 to September 2020, was reserved for testing its performance both on the same source (seen) catchments and on the target (unseen) catchments. This setup enables a more effective evaluation of the model's generalization capability and interpretability. Each catchment was represented by a combination of 15 static attributes and 11 hourly dynamic variables, as summarized in Table 1. These variables form the full input space were used for both forecasting and interpretability analyses.

Threshold discharges for flood event identification were established using return period flows derived from flood frequency analysis. Specifically, annual peak streamflow records from USGS gauge data (1965–2023) were modeled using a Pearson Type III distribution to estimate flows associated with 1- and 2-year return intervals. Although the 2-year return period is conventionally employed as a proxy for bankfull discharge and serves as a conservative design criterion, this study adopts the 1-year return period discharge to better support the objectives of flash flood early warning systems. The 1-year threshold provides both statistical rigor and operational relevance for detecting flash flood events. Table 2 summarizes the computed threshold discharges and associated hydrologic characteristics across both calibration (seen) and generalization (unseen) catchments. The discharge values are given as per unit catchment area.

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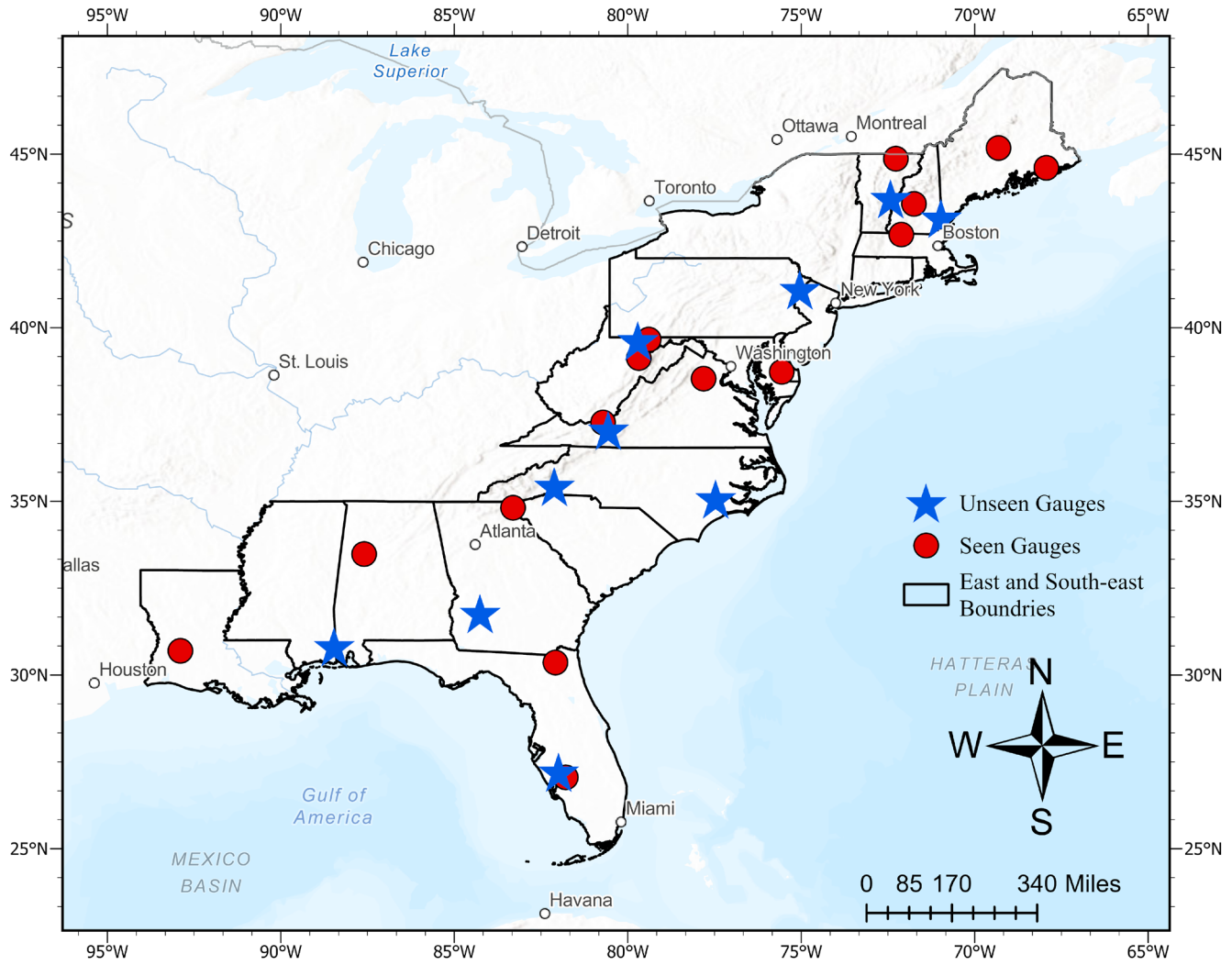


FIGURE 1 | Location of the study areas and gauging stations across the eastern and southeastern United States.

2.2 | TFT Model Description

The TFT is an attention-based deep neural network (DNN) designed for interpretable multi-horizon forecasting tasks (Lim et al. 2021). In contrast to fully autoregressive models such as transformer decoders—which generate step-by-step forecasts by reintroducing each predicted output into the model—TFT processes the full historical input sequence in parallel and produces forecasts for the entire prediction window at once. This structure allows for efficient parallel computation and enables the model to learn complex temporal patterns while maintaining strong forecasting accuracy over multiple time steps. The TFT architecture incorporates features such as dynamic variable selection, gating to filter out irrelevant information, and multi-head attention, which together enhance both prediction performance and interpretability (Lim et al. 2021). These strengths make TFT well-suited for hydrological forecasting, where understanding the interaction between static catchment attributes and time-varying meteorological inputs is key to making accurate predictions (Sadeghi Tabas et al. 2025).

The TFT is made up of several components that are designed to capture complex temporal relationships while improving the interpretability of the model (Lim et al. 2021). These components include variable selection networks (VSNs), gated residual networks (GRNs), a sequence-to-sequence encoder-decoder architecture, static covariate encoders, and a multi-head attention mechanism. Together, these parts allow the model to focus on the most relevant features and time steps to make accurate and explainable forecasts. GRNs help the model adjust its internal complexity depending on the nature of the input data (Lim et al. 2021). By selectively applying transformations and bypassing unnecessary operations, the model avoids overfitting and becomes more efficient. VSNs operate at each time step to identify which input features are most important. They assign attention-based weights to each variable, allowing the model to prioritize meaningful signals and suppress noise, which improves both performance and interpretability. Static covariate encoders are used to process catchment-specific characteristics such as climate, topography, and soil properties (Lim et al. 2021). These static attributes are embedded into context vectors and used throughout the model to condition the temporal representations, allowing TFT to adjust its forecasting behavior based on the physical and environmental context of each catchment. The model's temporal processing layers include a sequence-to-sequence LSTM encoder that captures short-term sequential patterns, as well as a multi-head attention mechanism that models long-term dependencies by learning which time steps are most

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TABLE 1 | Hourly catchment-averaged Maurer forcing data used as dynamic features as well as static features (time-invariant catchment attribute) including long-term average climatic and vegetation indices, as well as soil and topographical properties.

Type	No.	Category	Feature and unit	Description
Dynamic features	1	Meteorological	prcp (mm/day)	Precipitation
	2	Meteorological	PET (mm)	Potential evapotranspiration
	3	Meteorological	Humidity (kg/kg)	Specific humidity
	4	Meteorological	T (K)	Air temperature at 2 m above the surface
	5	Meteorological	Wind-U (m/s)	West to east wind (eastward positive, westward negative)
	6	Meteorological	Wind-V (m/s)	South to north wind (northward positive, southward negative)
	7	Meteorological	rlds (W/m^2)	Surface downward longwave radiation
	8	Meteorological	rsds (W/m^2)	Surface downward shortwave radiation
	9	Meteorological	Pressure (Pa)	Surface pressure
	10	Meteorological	convective_fraction	At 2 m above the surface
	11	Meteorological	potential_energy (CAPE) (J/kg)	Amount of energy available for convective process
Static features	1	Topographic	area_gages^2 (km^2)	Catchment area
	2	Topographic	slope_mean (m/km)	Catchment mean slope
	3	Topographic	elev_mean (m above sea level)	Catchment mean elevation
	4	Climate	p_mean (mm/day)	Mean daily precipitation
	5	Climate	pet_mean (mm/day)	Mean daily PET, estimated by N15 using Priestley–Taylor formulation calibrated for each catchment
	6	Climate	Aridity (–)	Aridity (PET/P, ratio of mean PET, estimated by N15 using Priestley–Taylor formulation calibrated for each catchment, to mean precipitation)
	7	Climate	p_seasonality (–)	Seasonality and timing of precipitation (estimated using sine curves to represent the annual temperature and precipitation cycles; positive (negative) values indicate that precipitation peaks in summer (winter); values close to 0 indicate uniform precipitation throughout the year)
	8	Climate	high_prec_freq (days/year)	Frequency of high precipitation days (≥ 5 times mean daily precipitation)
	9	Land cover	forest_frac (–)	Forest fraction
	10	Soil attributes	soil_porosity (–)	Volumetric porosity (saturated volumetric water content estimated using a multiple linear regression based on sand and clay fraction for the layers marked as USDA soil texture class and a default value (0.9) for layers marked as organic material; layers marked as water, bedrock, and "other" were excluded)

(Continues)

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TABLE 1 | (Continued)

Type	No.	Category	Feature and unit	Description
	11	Soil attributes	soil_conductivity (cm/h)	Saturated hydraulic conductivity (estimated using a multiple linear regression based on sand and clay fraction for the layers marked as USDA soil texture class and a default value (36 cm h^{-1}) for layers marked as organic material; layers marked as water, bedrock, and "other" were excluded)
	12	Soil attributes	max_water_ content (m)	Maximum water content (combination of porosity and soil_depth_statsgo; layers marked as water, bedrock, and "other" were excluded)
	13	Soil attributes	sand_frac (%)	Sand fraction (of the soil material smaller than 2 mm; layers marked as organic material, water, bedrock, and "other" were excluded)
	14	Soil attributes	silt_frac (%)	Silt fraction (of the soil material smaller than 2 mm; layers marked as organic material, water, bedrock, and "other" were excluded)
	15	Soil attributes	clay_frac (%)	Clay fraction (of the soil material smaller than 2 mm; layers marked as organic material, water, bedrock, and "other" were excluded)

influential for prediction. This combination helps the model effectively capture patterns that unfold over different time scales. Finally, the temporal fusion decoder integrates past observations, static catchment features, and known future inputs to produce multi-step probabilistic forecasts. These forecasts include uncertainty estimates, which are important for decision-making in applications like flood forecasting. The overall architecture and flow of information within TFT are illustrated in Figure 2, showing how dynamic and static features are processed through attention and feature selection to generate accurate and interpretable predictions (for further information about TFT structure please refer to Lim et al. 2021).

To tailor TFT for hourly streamflow forecasting, specific strategies for training and evaluation were employed, as detailed below. All input features were normalized as part of the data preprocessing pipeline prior to model training to ensure numerical stability and comparability across variables with different units and magnitudes. Continuous dynamic variables, including meteorological forcings and observed streamflow, were standardized using z-score normalization based on the mean and standard deviation computed from the training period only. The same normalization parameters were then applied for evaluation unseen datasets to prevent information leakage. Static catchment attributes were normalized across catchments using the same procedure. This normalization strategy follows established best practices for multivariate deep learning in hydrology and supports stable optimization, interpretability, and transferability of the trained TFT model across catchments with heterogeneous characteristics. TFT architecture offers a unique advantage for modeling hydrological processes across multiple catchments due to its capacity to jointly learn both global and local temporal patterns. Specifically, TFT integrates static covariates (catchment physical attributes) along with dynamic time series inputs, enabling the model to condition temporal predictions on the intrinsic physical attributes of each catchment. In this study, a single TFT model was trained across all catchments (15 seen catchments), allowing for the exploitation of shared temporal dependencies (global patterns), while simultaneously capturing catchment-specific hydrological responses (local patterns). This approach enhances the model's ability to generalize across heterogeneous catchments and improves predictive performance by leveraging both inter-catchment similarities and intra-catchment distinctions. As mentioned above, TFT follows an autoregressive encoder-decoder manner, where the model takes the first month of data to predict the next 24 h. In other words, validation was conducted using a sliding window comprising the preceding 30 days of observed data and a 24-h forecasting horizon. This approach promoted generalization across diverse hydrological conditions while mitigating overfitting and improving predictive accuracy (see Figure 3).

During the inference phase, similar sliding window mechanism was applied repeatedly across the entire test period (October 2014–September 2020) for both seen and unseen catchments. At each step, the 30-day historical input window was dynamically updated by incorporating the latest available observation and discarding the earliest, ensuring that forecasts consistently reflected the most recent information. This approach preserved the temporal integrity of the dataset and enabled the model to simulate realistic operational forecasting conditions by continually updating its predictions as new observations became available. Due to the semi-auto-regressive nature of the model, at each prediction step, TFT generates 24 forecasted discharge values corresponding to lead times from 1 to 24 h ahead. This forecasting structure enabled the construction of continuous hydrographs for different lead times, which is particularly beneficial for real-time flood forecasting and operational decision-making. Specifically, collecting all 1-h-ahead predictions from each time step produced a continuous hydrograph for the 1-h horizon, while extracting the second prediction across all time steps yielded a hydrograph for the 2-h horizon, and so on up to 24 h lead time. This methodology

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TABLE 2 | One- and two-year return period threshold values for discharge of 25 selected catchments.

Catchment no.		USGS no.	Drainage area (squared miles)	Location, state	1-year return period threshold (mm/h)	2-year return period threshold (mm/h)
Seen catchments	1	01022500	227	Narraguagus River at Cherryfield, ME	0.307	0.661
	2	01031500	296	Piscataquis River near Dover-Foxcroft, ME	0.424	1.337
	3	01078000	85.8	Smith River Near Bristol, NH	0.333	0.691
	4	01162500	19.4	Priest Brook Near Winchendon, MA	0.234	0.703
	5	01487000	75.4	Nanticoke River Near Bridgeville, DE	0.118	0.394
	6	01664000	619	Rappahannock River at Remington, VA	0.262	0.854
	7	02177000	207	Chattooga River Near Clayton, GA	0.391	1.443
	8	02231000	700	St. Marys River Near Macclenny, FL	0.098	0.279
	9	02298123	233	Prairie Creek Near Fort Ogden FL	0.313	0.335
	10	02464000	223	North River Near Samantha AL	0.475	1.516
	11	03069500	718	Cheat River Near Parsons, WV	0.832	1.588
	12	03076600	48.9	Bear Creek at Friendsville, MD	0.41	1.393
	13	03173000	299	Walker Creek at Bane, VA	0.286	0.975
	14	04296000	122	Black River at Coventry, VT	0.32	0.742
	15	08014500	510	Ouiska Chitto Creek Near Oberlin, LA	0.122	0.806
Unseen catchments	1	01073000	12.1	Oyster River Near Durham, NH	0.345	0.911
	2	01144000	690	White River at West Hartford, VT	0.233	0.976
	3	01439500	117	Bush Kill at Shoemakers, PA	0.27	0.769
	4	02092500	168	Trent River Near Trenton, NC	0.102	0.467
	5	02149000	79	Cove Creek Near Lake Lure, NC	0.206	1.74
	6	02297310	218	Horse Creek at SR 72 Near Arcadia, FL	0.049	0.366
	7	02350900	527	Kinchafoonee Creek at Pinewood, GA	0.069	0.239
	8	02479560	562	Escatawpa River Near Agricola, MS	0.185	0.726
	9	03070500	200	Big Sandy Creek at Rockvile, WV	0.686	1.321
	10	03170000	300	Little River at Graysontown, VA	0.246	0.796

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facilitated a detailed evaluation of predictive performance across both short and longer lead time horizons. By applying this structured evaluation across multiple catchments and forecast windows, the model's ability to generalize unseen hydrological patterns was rigorously assessed.

It's important to note that, during evaluation, the TFT model generated discharge forecasts for lead times of 1–24 h using past observed hydrological states and proxy future meteorological inputs constructed by shifting observed NLDAS reanalysis variables forward in time. This design isolates the forecasting capability of the TFT architecture under reanalysis forcing conditions and avoids confounding effects from numerical weather prediction uncertainty. The NWM v3.0 CONUS Retrospective Dataset (reanalysis), by contrast, represents a zero-lead hindcast driven by reanalysis meteorology; therefore, the comparison is intended as an operationally relevant benchmark rather than a strictly lead-time-equivalent forecast evaluation.

2.3 | National Water Model (NWM) Description

To forecast streamflow and related variables across the contiguous United States (CONUS), the National Water Model (NWM) Version 3.0 by NOAA (National Oceanic and Atmospheric Administration) is widely used. It is a distributed hydrologic modeling system with high resolution (typically finer than 1 km). The model incorporates hydrometeorological inputs—such as numerical weather forecasts, radar-based precipitation, soil moisture, snowpack, and land surface conditions—to drive its simulations (Cosgrove et al. 2024). Built upon the WRF-Hydro framework, the model represents surface and subsurface water processes across multiple spatial and temporal scales. It supports continuous operation and provides short-, medium-, and long-range forecasts, making it suitable for flood prediction, water resource management, and emergency response (Cohen et al. 2018).

Version 3.0 introduces significant enhancements over prior versions, including improved physical process representation, refined calibration methodologies, and expanded data assimilation capabilities. It also features an advanced ensemble forecasting system to quantify uncertainties in both meteorological forcings and hydrologic outputs. The model generates streamflow simulations for over 2.7 million river reaches, enabling fine-scale analysis of hydrologic dynamics.

NWM v3.0 CONUS Retrospective Dataset (reanalysis data) was used as a benchmark in this study for evaluating the performance of the TFT model. A custom Python script was developed to automate the retrieval and processing of historical NWM outputs from NOAA's public S3 data archive, enabling efficient extraction of streamflow time series at selected gauge locations. By comparing TFT predictions against NWM simulations, the study assesses the potential of data-driven models to achieve—or in some cases exceed—the forecasting skill of physics-based systems.

2.4 | Metrics for Performance Assessment

A set of performance metrics is used to evaluate both prediction accuracy and uncertainty. For accuracy, the Nash–Sutcliffe Efficiency (NSE) was used, where values closer to 1 indicate stronger agreement between simulated and observed streamflow. An NSE of 1 means a perfect match, 0 indicates performance equal to the observed mean, and values below 0 show worse performance. Additional error metrics included Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), with lower values reflecting better accuracy. To assess the model's ability to capture flood events peak magnitude and time to peak, Relative Peak Bias (RPB) and Peak Time Bias (PTB) were used. Positive or negative values of RPB and PTB indicate over- or under-estimation of peak flow and late or early peak time, respectively. For uncertainty evaluation, the P-factor, measuring the percentage of observations falling within the 95% prediction uncertainty (95PPU) band, and the R-factor, quantifying the average width of the uncertainty band relative to the standard deviation of observations, were used. Ideally, a good model has a high P-factor (approaching 1) and a low R-factor (approaching 0). The Total Uncertainty Index (TUI), combining both P-factor and R-factor, was also computed to provide an overall measure of uncertainty performance. The mathematical expressions for these metrics are presented in Table 3.

2.5 | TFT Model Training

The TFT model was trained using data from catchments included in the training set (15 seen catchments; from October 1997 to September 2014) and evaluated its generalization performance on both seen and unseen catchments during the test period (from October 2014 to September 2020). To achieve optimal model performance, Optuna was employed. Optuna is an advanced hyperparameter optimization framework that leverages Bayesian optimization with a Tree-structured Parzen Estimator (TPE) to efficiently explore high-dimensional parameter spaces (Akiba et al. 2019). One of Optuna's key advantages is its ability to dynamically prune unpromising trials based on intermediate training results, which is particularly valuable in deep learning, where model training is often computationally expensive. We conducted 30 Optuna trials, each representing a complete training and evaluation cycle with a distinct set of hyperparameters. Trials were allowed to run up to 25 epochs, but early stopping was applied with a patience of 5 epochs—training would halt early if the validation loss failed to improve for five consecutive epochs. This approach not only reduced computation time but also helped mitigate overfitting. To further accelerate the tuning process, the limit_train_batches parameter was set to 0.1, meaning only 10% of the training batches were used in each epoch during tuning. This was particularly important given the dataset's size, which included 17 years of hourly data and 26 input features per instance. Sampling a smaller portion of the data allowed for quicker evaluations without compromising the usefulness of the feedback.

It's important to note that TFT offers built-in interpretability that provides necessary understanding of role played by various input features in the model's predictions. The identification of the key features allows the model to decipher the pertinent pattern and relationship in the data, resulting in better accuracy and generalization of the model to new, unseen conditions/catchments. This allows the model to focus on relevant inputs and avoid noise, thus avoiding overfitting. TFT enhances

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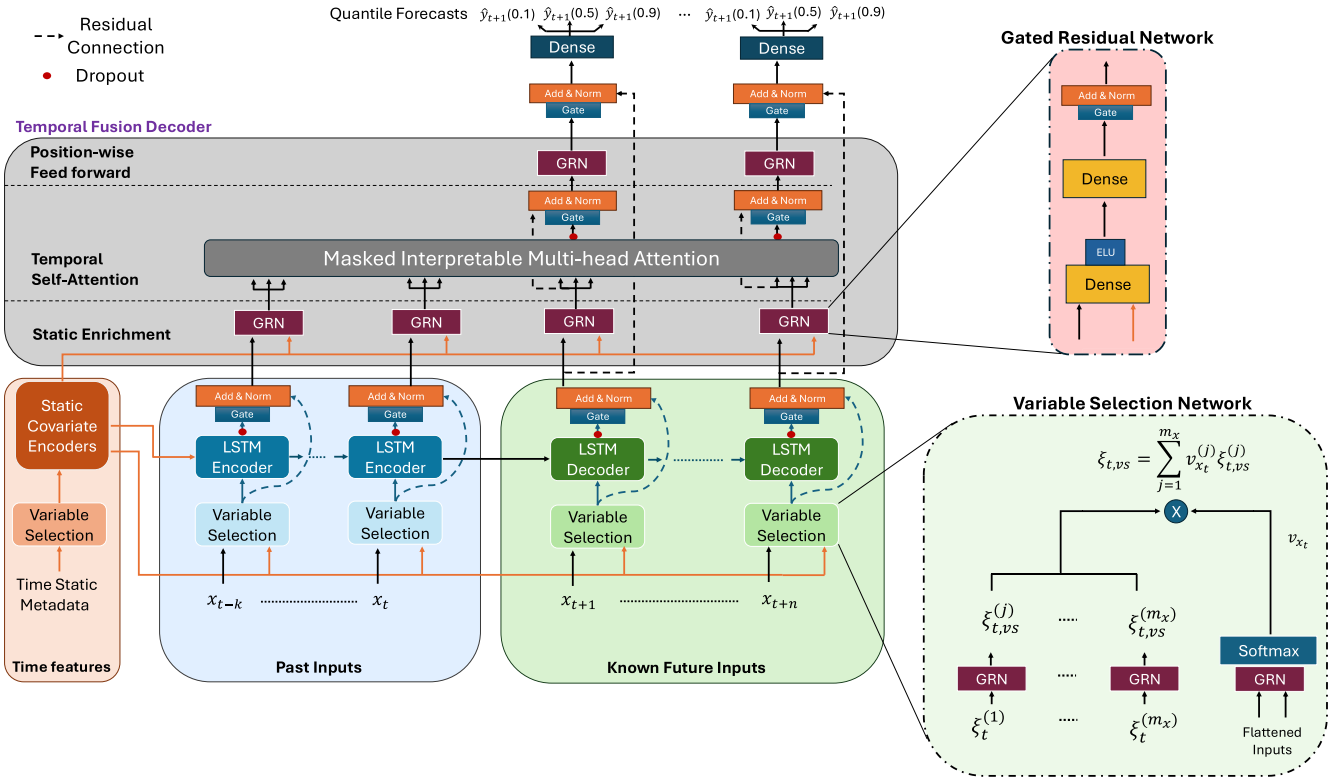


FIGURE 2 | Schematic overview of the TFT architecture for hydrological forecasting.

interpretability through separate VSNs. These VSNs handle static inputs, historical dynamic inputs, and known future inputs independently. The encoder processes historical time series data to generate temporal representations encapsulated in a context vector that captures the dynamics influencing future outcomes. The decoder integrates this context with known future inputs to produce forecasts across multiple time horizons. A key strength of the TFT lies in its ability to accommodate heterogeneous data types, enabling the model to effectively exploit a broad array of input features.

Table 4 summarizes the best hyperparameter configuration discovered during optimization. This configuration provides a strong balance between model accuracy and computational efficiency. While most hyperparameters were tuned automatically by Optuna, a batch size of 32 from the options [16, 32, 64, 128] was selected manually, based on initial testing and available GPU memory. We also fixed the total number of training epochs at 25 for consistency. The Adam optimizer was used for training given its widespread success and reliability in deep learning applications.

3 | Results and Discussion

3.1 | Interpretability Analysis of Input Features

To assess the interpretability of the model, variable-level attention contributions that are generalized for both the source (seen) and target (unseen) catchments are analyzed, encompassing static, historical dynamic, and known future variables (Figure 4). In addition, catchment-specific interpretability analyses for both static and future known inputs were conducted (see Section 3.2).

As illustrated in subplots (a) and (d) of Figure 4, the importance of static features differs between the seen and unseen catchments. This variation is expected, given that the two catchment groups are hydrologically distinct. What's particularly noteworthy is that the TFT model dynamically adapts its feature weighting during inference, despite never having seen the target catchments during training. This demonstrates the model's strong generalization capacity and supports its use in zero-shot I-TL setups. It highlights one of TFT's key strengths, its ability to identify and respond to differences in catchment characteristics during prediction, rather than being rigidly tied to training data behavior. The variable importance plots for the encoder inputs (subplots b and e) show that the model consistently ranks past discharge values as the most influential among all time-varying variables, highlighting their role as key hydrological state. For the future known inputs (subplots c and f), the model gives the highest priority to meteorological variables indicative of storm or rainfall conditions, such as total precipitation, convective fraction, and longwave radiation. These serve as critical meteorological forcing inputs, offering insights into upcoming weather patterns that influence runoff generation. This pattern holds true across both source (seen) and target (unseen) catchments, demonstrating the TFT's ability to effectively extract and leverage the most relevant hydrometeorological drivers, even when applied to previously unseen catchments.

In addition, the temporal attention profile provides empirical justification for the selected 30-day historical lookback window. As shown in Figure 5, although the TFT assigns the

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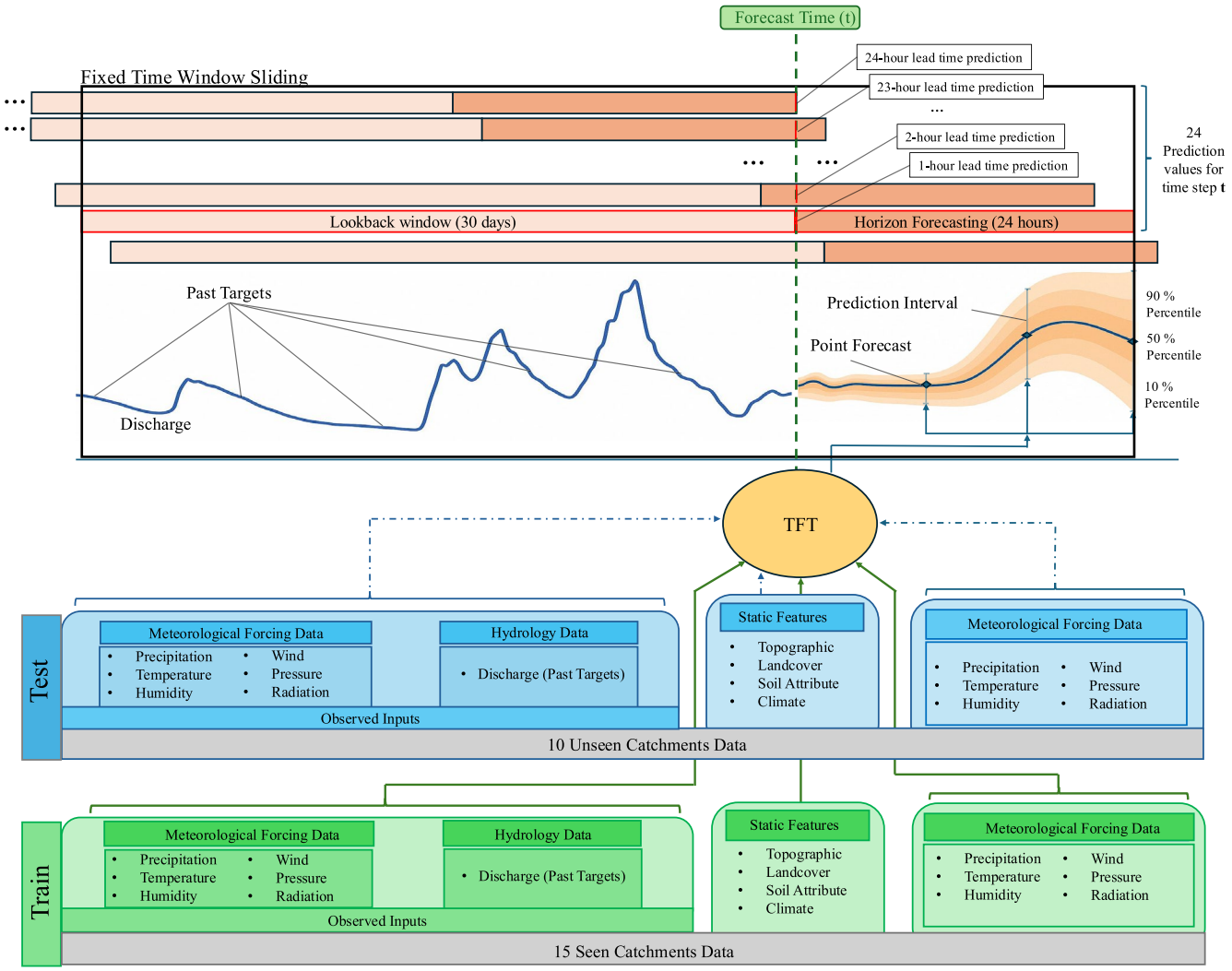


FIGURE 3 | Illustration of multi-horizon forecasting with known catchment static features, past time-dependent observed hydrological and meteorological forcing data, and a priori-known future time-dependent meteorological forcing data (e.g., forecasted).

highest attention weights to the most recent observations, it continues to attribute non-negligible attention to antecedent information extending back approximately 30 days (≈720 h). The smooth increase in attention toward the forecast time indicates that the model integrates both short-term dynamics and longer-term hydrological memory, reflecting the influence of antecedent moisture conditions, groundwater interactions, and seasonal wetness on streamflow generation. This behavior suggests that shorter input windows would truncate informative signals, while longer windows would yield diminishing returns as attention weights beyond 30 days approach negligible values.

3.2 | Interpretability Results for Individual Catchment

The spatial heterogeneity of hydrological drivers across various catchments can be assessed by exploring the interpretability analysis of the TFT. The quantitative role of the feature importance in minimizing the prediction error at the catchment level identifies which static and dynamic variables have the most relative contribution for streamflow forecasting under diverse hydroclimatic conditions. Thus, enhancing the understanding of the localized hydrological behavior. In addition, it provides transparency of the DL architecture. The results of the feature importance have practical implications for guiding future data acquisition strategies, refining feature selection methodologies, and uncovering region-specific forecasting challenges. Beyond predictive performance, the interpretability analysis of the DL models enables more scientifically grounded and operationally relevant applications. The attention scores produced by the TFT were multiplied by 100, thereby expressing feature importance in percentile terms to facilitate clearer comparison across locations. It is important to note that this analysis was applied to both static features and dynamic future-known inputs (decoder variables) with results presented in Figures 6 and 7, respectively. The sum of the feature importance values equals 100% for each catchment and reflects allocation of attention across features. The scores are relative within each catchment as well as across the catchment groups (seen and unseen), providing comparative utility of each feature under specific spatial contexts.

It is important to note that TFT does not differentiate between positive and negative impacts of variables on streamflow. Instead, it highlights those features that are most informative for minimizing predictive error, features that exhibit consistent

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TABLE 3 | Performance metrics for model evaluation, including deterministic accuracy measures and uncertainty quantification indicators.

Metric's category	Metric	Formula	Equation no.
Performance evaluation metrics	NSE	$1-\frac{\sum_{i=1}^{n}\left(Q_{\mathrm{obs},i}-Q_{\mathrm{sim},i}\right)^2}{\sum_{i=1}^{n}\left(Q_{\mathrm{obs},i}-\overline{Q}_{\mathrm{obs}}\right)^2}$	(1)
	RMSE	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(Q_{\mathrm{obs},i}-Q_{\mathrm{sim},i}\right)^2}$	(2)
	MAE	$\frac{1}{n}\sum_{i=1}^{n}\left\|Q_{\mathrm{obs},i}-Q_{\mathrm{sim},i}\right\|$	(3)
	RPB	$\mathrm{RPB}=\frac{Q_{\mathrm{peak}}^{\mathrm{sim}}-Q_{\mathrm{peak}}^{\mathrm{obs}}}{Q_{\mathrm{peak}}^{\mathrm{obs}}}$	(4)
	PTB	$\mathrm{PTB}=t_{\mathrm{peak}}^{\mathrm{sim}}-t_{\mathrm{peak}}^{\mathrm{obs}}$	(5)
Uncertainty assessment metrics	P-factor	$\frac{\text{Number of Obs. inside bounds}}{\text{Total number of Obs.}}$	(6)
	R-factor	$\frac{1}{n}\sum_{i=1}^{n}\frac{\text{Upper Bound}-\text{Lower Bound}}{\mathrm{Std}\left(Q_{\mathrm{obs},i}\right)}$	(7)
	TUI	$\frac{P-\text{Factor}}{R-\text{Factor}}$	(8)

Note: $Q_{\mathrm{obs},i}$: observed discharge at time step i; $Q_{\mathrm{sim},i}$: simulated discharge at time step i; $\overline{Q}_{\mathrm{obs}}$: average of observed discharge; n: number of time steps; upper bound and lower bound: prediction interval limits; Std: standard deviation of observed discharge.

TABLE 4 | TFT's optimal set of hyperparameters identified through Optuna tuning process.

Hyperparameter	Search space	Optimum	Description
Learning rate	0.0001–0.1	0.001	Controlling the learning speed of the model
Hidden layer size	8–128	24	The number of nodes in the model's hidden layers, affecting its capacity to learn complex representations
Dropout rate	0.1–0.5	0.2982	The probability of randomly dropping out a neuron during training, used as a regularization technique to prevent overfitting
Number of attention heads	1–4	3	Specifies how many parallel attention mechanisms are used in the multi-head attention layer, enabling the model to focus on different representation subspaces
Gradient clipping value	0.01–1.0	0.0413	Limits the gradient norm during backpropagation to prevent exploding gradients, ensuring training stability
Hidden continuous state size	8–128	15	Within the model's feature transformation block, influencing the capacity to represent complex input patterns

statistical relationships with streamflow, regardless of whether they increase or decrease it. For example, if variables such as "clay fraction" or "soil porosity" show regionally strong correlations with streamflow behavior, regardless of the underlying physical processes or the direction of influence, TFT will assign them high importance based on their predictive utility. This interpretability approach is purely data-driven, focusing on explanatory power rather than causal direction. This characteristic of TFT is clearly reflected in our results: Figure 6 shows the most influential static feature identified by the TFT model for each catchment. Red-circled markers represent unseen catchments, while dotted ovals group neighboring catchments from source and target groups (seen and unseen) that share similar (conceptually and correlatively) dominant features. This spatial consistency highlights the TFT model's ability to detect and utilize features that provide the most informative signals for accurate prediction, even in catchments that were entirely unseen during training. As an example, in Vermont, the target (unseen) catchment was most influenced by *clay-fraction*, while the nearby source (seen) catchment emphasized *soil-porosity*, two features that, although distinct, are both tied to the soil's ability to retain and transmit water. Similarly, along the Maryland–Pennsylvania border, the model selected comparable features for neighboring catchments,

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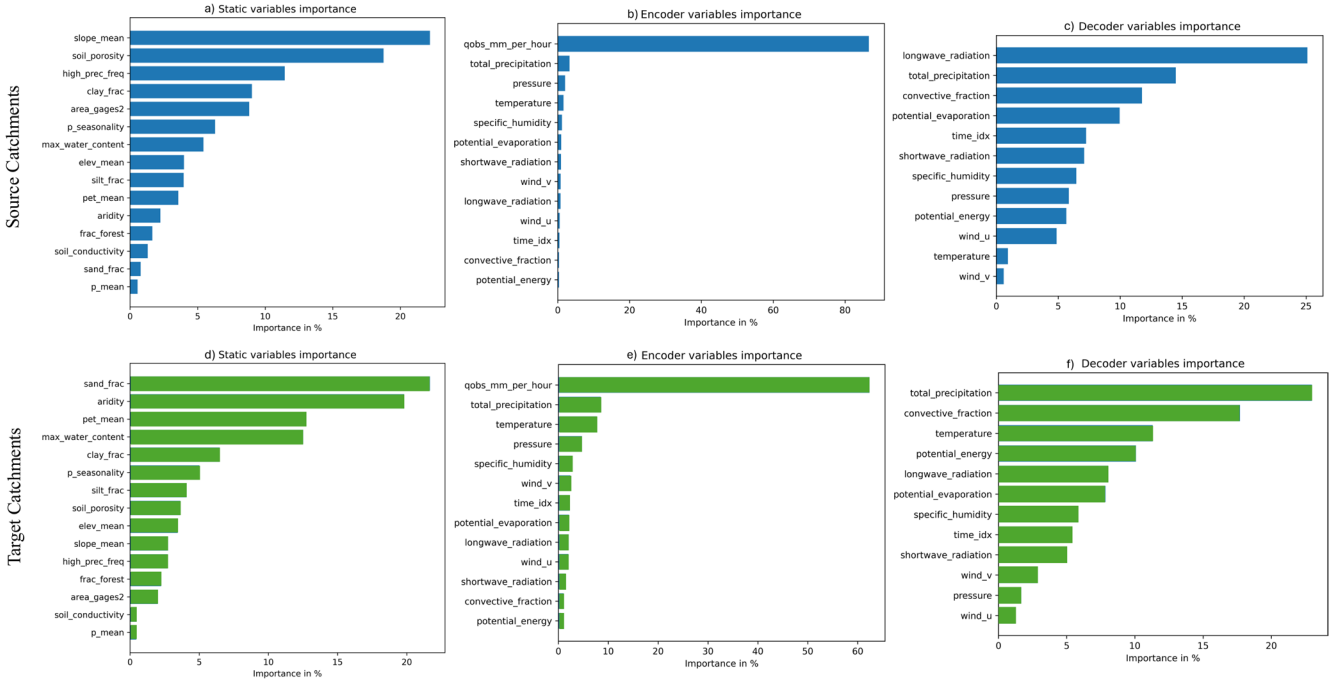


FIGURE 4 | Variable importance from the trained TFT model for source (top) and unseen (bottom) catchments: (a, d) static features, (b, e) encoder variables (past inputs), and (c, f) decoder variables (future known inputs). Importance reflects each feature's relative contribution to the model's predictions.

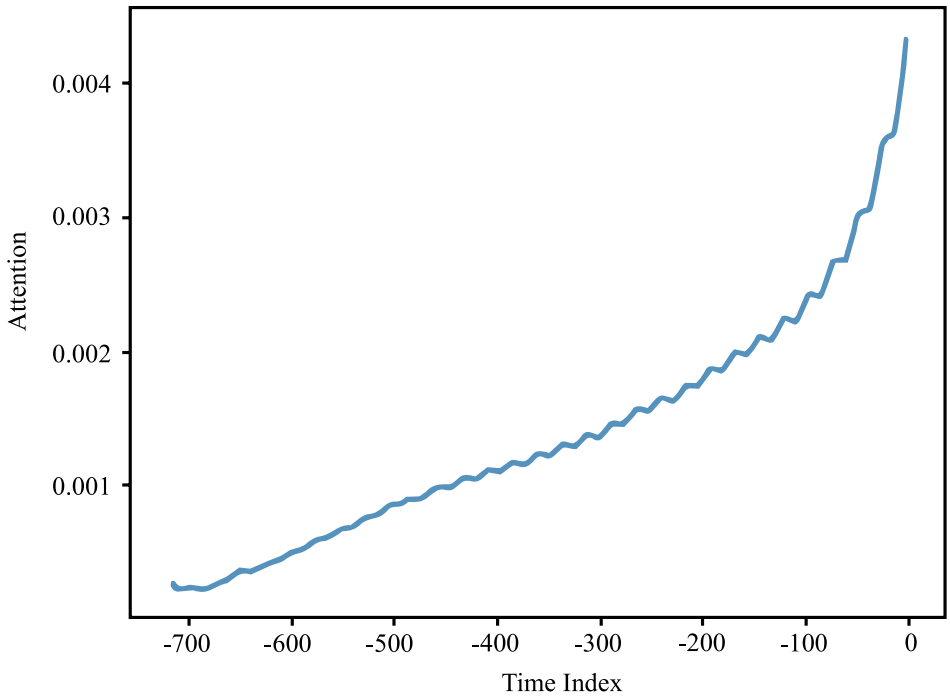


FIGURE 5 | TFT temporal attention weights over the 30-day (720-h) lookback window.

indicating that it captured regional soil behavior effectively. In Virginia, soil-related variables once again emerged as dominant, suggesting that subsurface controls were more predictive of streamflow than climatic or topographic factors in that region. A particularly notable case appears in northern part of South Carolina, where TFT assigned high importance to *soil-porosity* in the source catchment and to *maximum-water-content* in the neighboring target catchment, attributes that are hydrologically linked through their influence on infiltration, water retention, and streamflow response. In Florida, TFT identified *soil-conductivity* and *slope-mean* as the most influential static features. While these features do not necessarily cause increased streamflow due to Florida's characteristically low slopes and sandy soils, their consistent relationship with observed streamflow patterns in the region made them particularly useful to the model. TFT likely leveraged the

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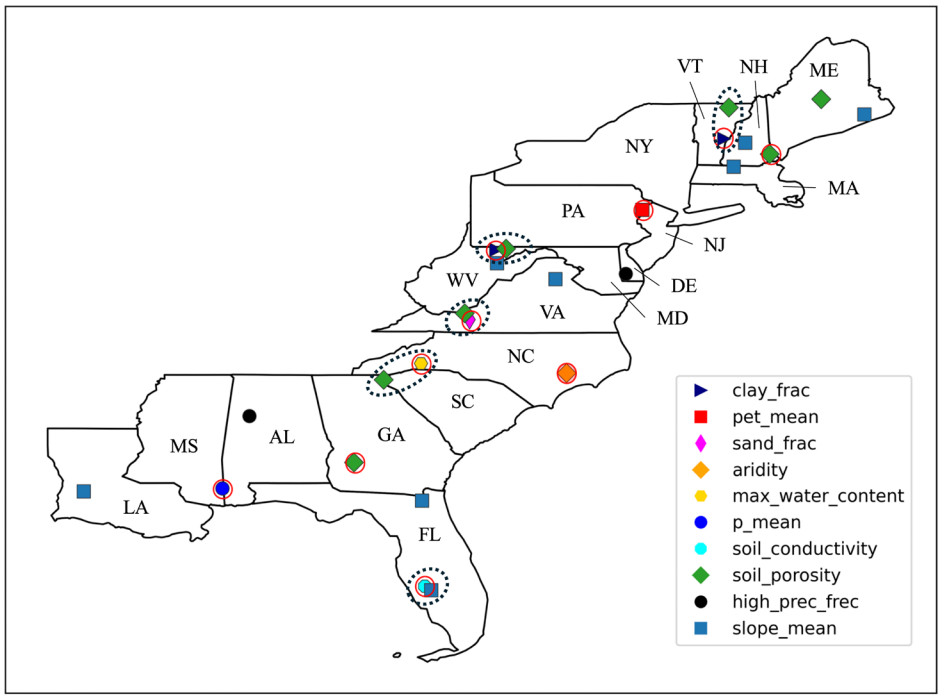


FIGURE 6 | Most important catchment static feature derived from TFT.

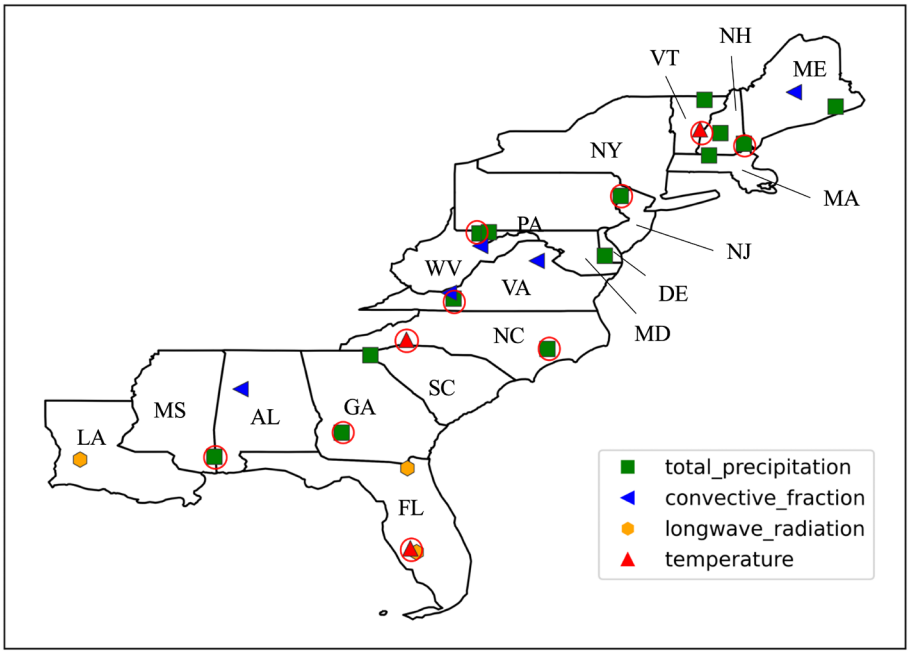


FIGURE 7 | Most important catchment dynamic feature derived from TFT decoder.

regional variability in slope and conductivity as informative signals, helping to distinguish the streamflow behavior of Florida catchments from others, regardless of whether those features drive slow or rapid streamflow processes in a physically causal sense. It is important to note that feature importance results derived from the TFT should be interpreted as performance-based indicators of how the model utilizes available information, rather than as evidence of causal relationships or definitive validation of underlying physical processes, particularly in the presence of correlated predictors.

Figure 7 illustrates the most influential future dynamic (decoder) features identified by the TFT model for each catchment. Across most catchments, total precipitation emerges as the dominant driver of streamflow forecasts, underscoring its central role in flow generation. However, in several specific regions, the model highlights convective fraction as the most informative feature, suggesting that localized storm activity and atmospheric instability significantly influence hydrologic response in those areas. Notably, in the southern catchments, features related to temperature and longwave radiation gain

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TABLE 5 | Performance assessment of the TFT model (24-h lead time forecasts) for hourly streamflow forecasting across unseen catchments during the test period from October 2014 to September 2020.

	Performance metrics						Uncertainty metrics		
	NSE		RMSE		MAE		R-factor	P-factor	TUI
USGS no.	NWM	TFT	NWM	TFT	NWM	TFT	TFT	TFT	TFT
01073000	0.406	0.738	0.062	0.041	0.033	0.016	0.725	0.918	1.267
01144000	0.217	0.743	0.071	0.041	0.034	0.015	0.888	0.908	1.022
01439500	0.577	0.829	0.058	0.037	0.035	0.016	0.796	0.849	1.067
02092500	0.473	0.194	0.204	0.252	0.051	0.025	0.197	0.879	4.464
02149000	0.655	0.655	0.071	0.071	0.030	0.022	0.793	0.906	1.143
02297310	0.687	0.835	0.048	0.035	0.021	0.010	0.569	0.971	1.706
02350900	0.605	0.582	0.033	0.034	0.018	0.010	0.539	0.882	1.638
02479560	0.811	0.614	0.058	0.083	0.032	0.022	0.475	0.882	1.859
03070500	0.705	0.761	0.067	0.061	0.033	0.022	0.737	0.934	1.267
03170000	0.663	0.597	0.052	0.057	0.020	0.014	0.693	0.926	1.338
Mean	0.580	0.655	0.072	0.071	0.031	0.017	0.641	0.910	1.677

greater importance, indicating the enhanced role of energy inputs in shaping streamflow dynamics under warmer, more humid climatic conditions. This spatial variability in feature importance demonstrates TFT's ability to adaptively prioritize the most relevant drivers based on regional hydroclimatic characteristics.

3.3 | Quantitative Performance Assessment

The section provides a quantitative assessment of how well the TFT model reproduces the observed streamflow across the studied catchments. The assessment is based on performance metrics (NSE, RMSE, and MAE) and uncertainty metrics (R-Factor, P-Factor, and TUI). Results for both continuous streamflow hydrograph simulation and event-based flood forecasting are presented.

3.3.1 | Assessment of Continuous Hydrograph Simulation

Table 5 presents a comprehensive evaluation of the TFT model's performance. The performance and uncertainty metrics are for 24-h lead time hourly streamflow forecasting across target (unseen) catchments, benchmarked against the NWMv3.0. The results demonstrate that TFT consistently outperforms NWM in terms of both predictive accuracy and uncertainty quantification, even though it was not trained for these catchments. On average, TFT achieves a higher NSE of 0.655 compared to 0.580 for NWM, indicating improved skill in reproducing observed streamflow. RMSE and MAE values are also slightly lower with TFT (0.071 and 0.017, respectively) than with NWM (0.072 and 0.031), reflecting better error minimization across catchments. It's important to note that there was one catchment for which TFT underperformed relative to NWM (USGS no. 02092500, Trent River near Trenton, NC, a coastal plain basin in eastern North Carolina). This catchment exhibits hydrological characteristics that differ markedly from many of the catchments used for model training, including very low topographic relief, shallow groundwater tables, strong groundwater–surface water coupling, and prolonged wetness conditions. Streamflow dynamics in this catchment are often storage-controlled, with slow recession behavior but with rapid responses to rainfall. Because the TFT model was not trained on a relatively close catchment to this, this catchment likely represents a distributional shift in both physiographic attributes and runoff generation mechanisms relative to the training data. As a result, the learned relationships may not transfer optimally in a zero-shot setting. In contrast, the physics-based NWM explicitly represents groundwater storage, channel routing, and backwater effects, which can better accommodate the dominant processes in coastal plain systems, explaining its relatively stronger performance at this site.

The quantile loss function in the TFT framework was used to quantify predictive uncertainty. The quantile loss function allows the model to learn conditional quantiles of the target distribution, thus generating full probabilistic forecasts as opposed to point estimates generated by traditional loss functions. The quantile function in the TFT model was used to estimate the 2.5th and 97.5th percentiles of streamflow, forming a 95% prediction uncertainty interval (95PPU) at each time step. The uncertainty metrics reported in Table 5 highlight the robustness and calibration of the model. The P-Factor, which indicates the proportion of observed values falling within the predicted intervals, averaged 0.910 across catchments, showing that the TFT captured over 91% of observed values within its forecast bounds. Meanwhile, the R-Factor, reflecting the average width of the uncertainty band relative to the variability in observations, averaged 0.641, suggesting that the intervals were appropriately narrow and not excessively conservative. The Total TUI, which combines both coverage and sharpness, had a mean value of

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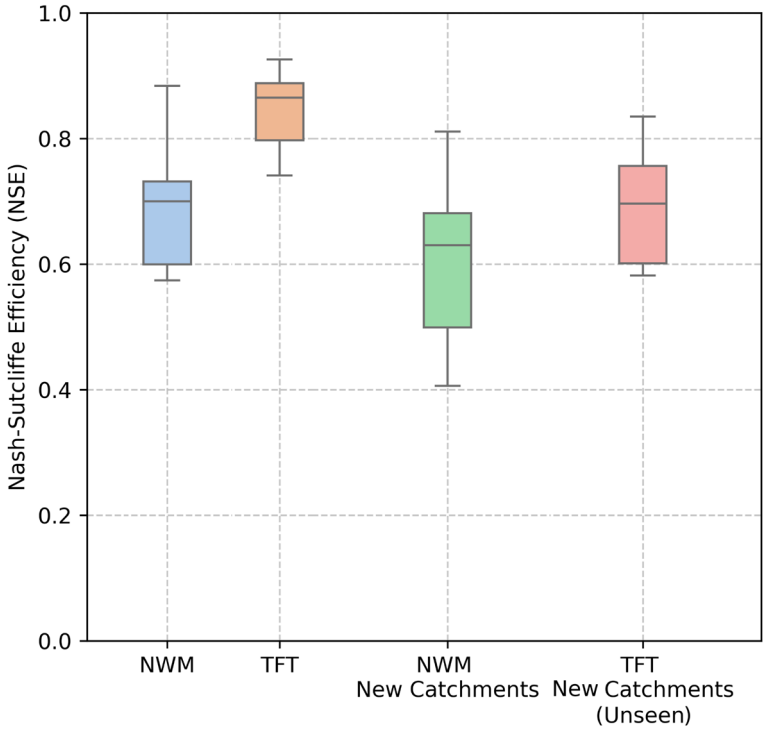


FIGURE 8 | Boxplots of NSE scores comparing TFT performance on seen and unseen catchments versus NWMv3.0 reanalysis.

1.677, indicating a favorable trade-off between reliability and informativeness.

Figure 8 presents the boxplots comparing the NSE distribution for TFT and NWM models across both source (seen) and target (unseen) catchments. The first two boxes (left side) represent performance on source (seen by TFT) catchments, where both models were evaluated on the catchments that were also used for training or calibration. TFT shows significantly better performance, with a higher median NSE, tighter interquartile range (IQR), and fewer low outliers compared to NWM. This indicates that TFT not only achieves more accurate streamflow predictions but also demonstrates more consistent performance across the source catchments. The right two boxes depict model performance on target catchments (TFT model had never seen the data related to these catchments, but NWM model was calibrated on those catchments). Again, TFT outperforms NWM, achieving a higher median NSE and a narrower spread. While both models experience some degradation in predictive skill compared to the source catchments, TFT maintains stronger generalization in this zero-shot transfer learning setting. In contrast, NWM shows a broader range and lower median NSE values, indicating more variability and weaker reliability. Overall, this figure reinforces the superior accuracy, robustness, and transferability of the TFT model. It demonstrates that TFT is not only effective when applied to familiar catchments but also generalizes well to unseen regions, offering a valuable advantage for streamflow forecasting in data-scarce environments.

To evaluate the TFT model's ability to quantify predictive uncertainty, Figure 9 presents median forecasts (red lines) alongside observed discharges (blue lines) for five representative unseen catchments, accompanied by grey-shaded regions denoting the 95PPU intervals. The close correspondence between predicted and observed flows across all stations, both in timing and magnitude, demonstrates the model's strong capacity to generalize temporal dynamics across diverse hydrological regimes. This performance is especially significant considering the zero-shot I-TL setting, where the model was applied to completely unseen catchments without any training or fine-tuning. The TFT model effectively tracks both baseflow and high-flow conditions, including the sharp rise and recession limbs of discharge peaks. Its ability to tightly follow the observed peaks, even during complex hydrological periods, reflects a well-calibrated forecasting mechanism. The uncertainty bands further support this; during periods of stable or low flow, the intervals remain narrow and responsive, avoiding unnecessary inflation of uncertainty. In contrast, during flood events or high variability, the intervals appropriately widen, capturing the increased uncertainty and complexity associated with extreme conditions. This dynamic behavior of the prediction intervals indicates the model's adaptability and nuanced understanding of varying flow regimes. In several catchments, even where gaps in observed data are present, the TFT produces forecasts that resume smoothly and closely match the observed trends after the missing periods. This reinforces the model's robustness in handling incomplete data scenarios, a valuable trait for real-world applications. To summarize, these results highlight several of the TFT's strengths: precise and adaptive uncertainty quantification, high predictive accuracy for short-term streamflow peaks, and resilience in the face of sparse observations. The consistent containment of observed flows within the uncertainty bands across different regimes suggests that the model strikes a strong balance between coverage (high P-factor) and informativeness (moderate to low R-factor), making it a reliable and accurate tool for short-term flood forecasting in new or data-limited catchments.

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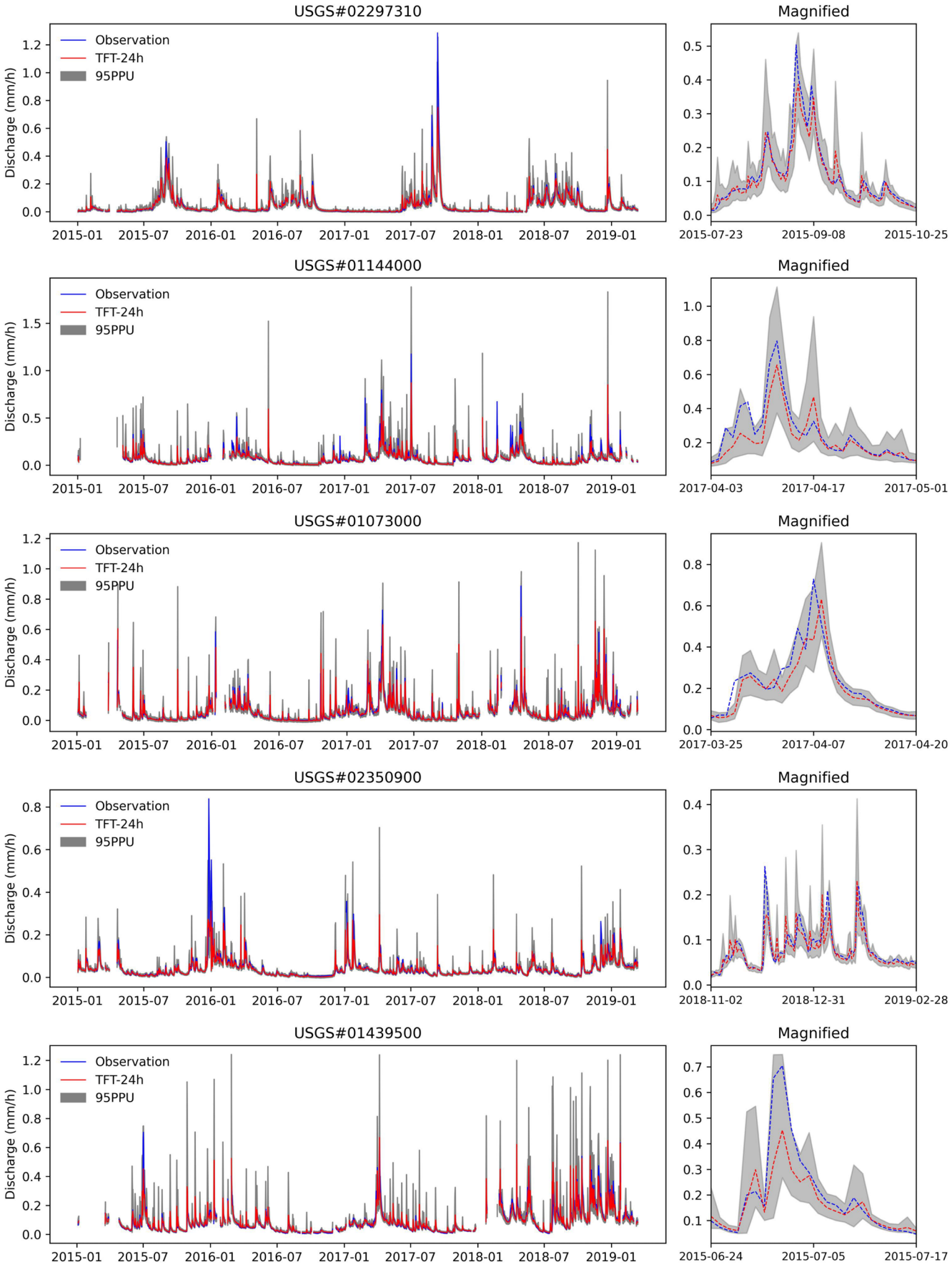


FIGURE 9 | Legend on next page.

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FIGURE 9 | Observed and simulated streamflow for five representative USGS catchments during the 2015–2019 test period using the TFT model (model never seen these catchments during the training). In each plot, the shaded region represents the 95PPU, bounded by the 2.5th and 97.5th percentiles predicted by the TFT model. The observed discharge is overlaid for reference. Across all five catchments, the observed values predominantly fall within the 95PPU bounds, confirming the model's ability to characterize forecast uncertainty accurately. Zoomed-in panels on the right highlight sample flood events to demonstrate the model's effectiveness in reproducing peak discharges and quantifying forecast uncertainty.

3.3.2 | Event-Based Performance Assessment Across Various Lead Times

To assess the performance of the TFT model in forecasting under extreme hydrological conditions, we carried out a targeted event-based evaluation focused specifically on short-term flood forecasting. This assessment involved 10 distinct flood events, each drawn from a different unseen catchment, and selected based on exceeding the 1-year return period threshold, ensuring that the events represent hydrologically significant and potentially hazardous conditions. By isolating high-impact events across diverse geographical and climatic regions not included in the training set, this evaluation provides a rigorous test of TFT's ability to generalize and respond accurately in operationally challenging, data-scarce environments. The selected events and corresponding forecasts are illustrated in Figure 10, which shows the comparative model performance across a range of lead times and hydrological behaviors. Each event was forecasted using the TFT model at five lead times (1, 6, 12, 18, and 24 h) to assess the model's short-range predictive capability across a range of operational horizons. These forecasts are directly compared to the corresponding reanalysis data from the NWMv3.0 to benchmark performance against an established physics-based system. Unlike general time-series evaluations, this event-focused approach directly reflects the model's utility in real-world applications such as emergency response, early warning systems, and flood preparedness, where both timing and accuracy of forecasts are critical. The flood threshold used to identify each event is indicated by a dashed horizontal grey line in each subplot of Figure 10, marking the critical discharge levels associated with the return-period-based benchmark. In addition, the selected events vary in flood structure, ranging from sharp single peaks to complex multi-peak patterns, allowing for a robust evaluation of the model's flexibility and responsiveness under different hydrological scenarios. This diversity further reinforces confidence in the TFT's capability as a reliable, data-driven alternative for flood forecasting across a broad range of catchment types and flood regimes.

As seen in Figure 10, the TFT model captures both the shape and timing of flood hydrographs with high fidelity across all forecast lead times. The 1-h lead time forecasts closely mirror observed discharges, with nearly perfect alignment during both the rising and recession limbs, particularly in sharply peaked events such as those at USGS no. 02149000 and USGS no. 03070500, where the flood hydrograph is smooth and well-defined. These cases highlight the model's ability to rapidly respond to incoming flood signals with minimal temporal lag. Even as the lead time increases to 6, 12, 18, and 24 h, the model maintains a consistently strong level of accuracy. The general hydrograph structure remains well preserved, and the timing of peaks remains closely synchronized with observations. Notably, in more complex flood events, such as those in catchments USGS no. 02297310 and USGS no. 02350900, which exhibit multi-peak or prolonged response patterns, the TFT model continues to deliver temporally stable and responsive forecasts. This reflects the model's capacity to retain useful temporal information and respond to varied runoff dynamics even in challenging scenarios. Further, in hydrologically diverse catchments such as USGS no. 02149000 and USGS no. 01073000, the TFT model demonstrates its ability to adapt to distinct flood response behaviors. In USGS no. 02149000, the event is characterized by a sharp, high-magnitude flood peak followed by a relatively quick recession, indicating a fast-responding catchment. Conversely, USGS no. 01073000 exhibits a more gradual rise and fall, with a broader hydrograph and delayed peak, suggesting a catchment with greater storage, flatter terrain, or more permeable soils, resulting in a slower hydrologic response. Despite these contrasting dynamics, the TFT model successfully adjusts to each pattern, capturing both the rapid spike in the flashy catchment and the more drawn-out flood wave in the slower-responding catchment. These examples highlight the model's flexibility in generalizing to varying runoff generation mechanisms, further underscoring its suitability for real-time forecasting across a spectrum of hydrological conditions.

In addition, the TFT outperforms NWMv3.0 in both the accuracy and adaptability of predictions. The NWM reanalysis often shows lagged responses, overestimated or underestimated peaks, or smeared hydrographs, especially during rapid rise events (e.g., USGS no. 02092500 and USGS no. 03070500). In contrast, TFT predictions are more responsive and closely follow the observed peaks, reinforcing the advantage of data-driven approaches for short-term flood forecasting in dynamic catchments. Overall, these results underscore TFT's robustness across a wide range of flood morphologies, including short, intense events, multi-day floods, and complex multi-peak scenarios, across all lead times. This positions the model as a highly reliable tool for flood forecasting applications in operational contexts.

The quantitative results (summarized in Table 6) further validate superior performance. At a 1-h lead time, the model achieves NSE scores exceeding 0.95 across nearly all catchments, with some cases reaching up to 0.982, underscoring the model's exceptional accuracy in short-term streamflow forecasting. While a gradual reduction in performance is observed with increasing lead time, as expected in any forecasting system, NSE values remain consistently high even at 24-h forecasts, with most catchments maintaining scores above 0.74. For instance, catchment 02297310 achieves an NSE of 0.86 at the 24-h horizon, dramatically outperforming the NWM reanalysis, which scored just 0.536. TFT also excels in bias-related and timing metrics, offering substantial improvements over the NWM. The RPB remains close to zero across all lead times, indicating a strong ability to capture the correct flood magnitude. In contrast, the NWM often shows considerable peak underestimation, e.g., an RPB of −0.796 in catchment 01144000.

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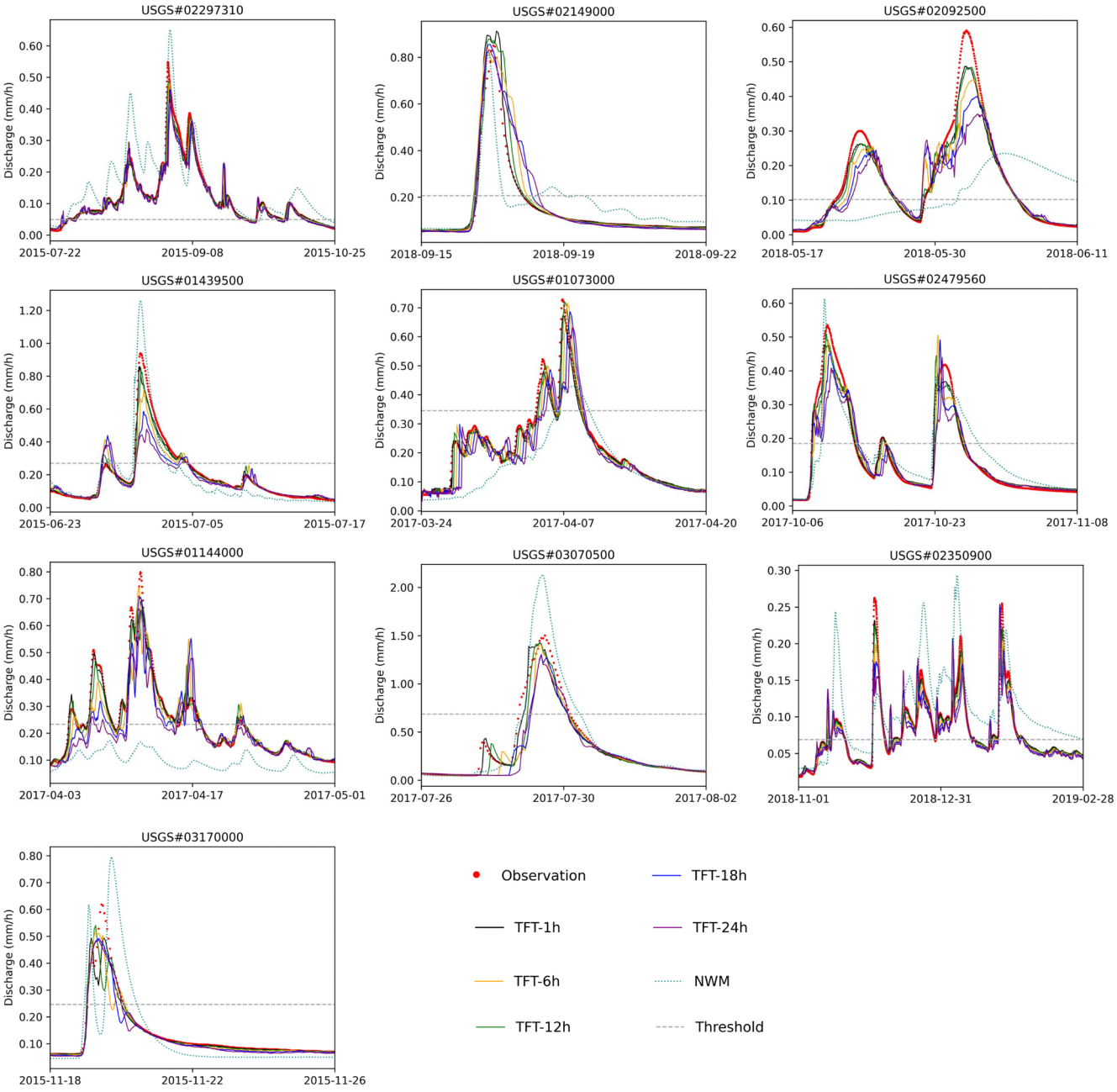


FIGURE 10 | Event-based TFT performance assessment across various lead-times (1–24 h) along with NWMv3.0 reanalysis and observations.

The model also demonstrates enhanced temporal accuracy in predicting peak flow timing, as measured by the PTB. In catchment 02297310, for example, the TFT shows a PTB of just 2 h at a 24-h lead time, while the NWM exhibits a much larger delay of 6 h. These improvements in both magnitude and timing underscore TFT's effectiveness in delivering timely and reliable flood forecasts. These findings are reinforced by the hydrographs in Figure 10, where TFT consistently provides tighter, more synchronized forecasts across different flow regimes. Compared to the NWM, which often shows delayed or suppressed peak responses, TFT outputs are more responsive—key traits for operational early warning systems. Overall, the results demonstrate that TFT is not only accurate in the short term but also scalable across time horizons, proving itself to be a robust and reliable solution for short-term flood forecasting, even in data-limited and hydrologically diverse settings.

4 | Conclusion

This study demonstrates the effectiveness of the TFT in hourly forecasting of streamflow in catchments, where no data from the target catchments were used during model training. By framing the problem as a true zero-shot I-TL task, we evaluated TFT's ability to generalize across catchments using meteorological forcing data and gauging station inputs as well as static catchment attributes during inference. Despite never being exposed to these catchments during training, TFT achieved strong predictive performance, consistently outperforming the NWMv3.0 across key metrics such as NSE, peak timing, and flood magnitude accuracy. One of TFT's core strengths lies in its ability to dynamically select and weight relevant features at inference time. Through its static covariate encoder and variable selection networks, the model adapts to diverse hydrological settings without retraining/

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TABLE 6 | Comparison of TFT forecasts for 15 sample events over various lead times against observation and NWMv3.0 reanalysis.

Metrics	NSE						RPB						PTB					
Model	TFT (lead time; h)					NWM	TFT (lead time; h)					NWM	TFT (lead time; h)					NWM
USGS no.	1	6	12	18	24	Reanalysis	1	6	12	18	24	Reanalysis	1	6	12	18	24	Reanalysis
01073000	0.980	0.944	0.860	0.792	0.745	0.465	−0.074	−0.074	−0.011	−0.018	−0.058	−0.132	1	2	2	3	3	0
01144000	0.979	0.920	0.839	0.795	0.722	−0.796	−0.134	−0.164	−0.069	−0.115	−0.180	−0.785	−1	−2	−3	−3	−3	−5
01439500	0.975	0.953	0.864	0.757	0.666	0.797	−0.086	−0.096	−0.234	−0.378	−0.492	0.344	2	2	3	4	4	1
02092500	0.962	0.953	0.892	0.797	0.684	−0.254	−0.175	−0.180	−0.240	−0.323	−0.409	−0.602	4	4	4	5	5	8
02149000	0.945	0.891	0.831	0.859	0.874	0.721	0.073	0.033	−0.003	0.007	−0.022	−0.034	−2	−3	−3	−2	−3	−3
02297310	0.982	0.979	0.947	0.905	0.860	0.536	−0.136	−0.115	−0.123	−0.161	−0.214	0.186	1	1	2	2	2	6
02350900	0.978	0.957	0.876	0.786	0.705	−0.038	−0.117	−0.143	−0.107	−0.034	−0.080	0.117	2	4	5	5	6	5
02479560	0.979	0.960	0.915	0.867	0.806	0.821	−0.110	−0.047	−0.057	−0.083	−0.180	0.144	−3	3	4	4	5	−4
03070500	0.951	0.905	0.841	0.846	0.836	0.807	−0.050	−0.056	−0.067	−0.132	−0.136	0.422	−1	−2	−2	−3	−3	−1
03170000	0.925	0.854	0.894	0.936	0.941	0.043	−0.202	−0.125	−0.163	−0.204	−0.214	0.287	−1	−1	−2	−2	−2	5

fine-tuning, enabling it to identify regionally dominant variables in minimizing prediction error, even in previously unseen catchments. This capability was clearly reflected in the consistent and hydrologically meaningful feature importance patterns observed across catchments, and in the model's skill in replicating both single- and multi-peak flood events with high fidelity. As discussed, TFT also produces probabilistic forecasts, offering valuable uncertainty quantification alongside its predictions—a crucial aspect for operational decision-making in flood risk management. It is important to note that TFT is an autoregressive model, relying on recent discharge observations as input for forecasting. As such, it cannot be applied to completely ungauged catchments where no streamflow data are available. Instead, its strength lies in partially gauged, newly gauged or data-scarce catchments, where limited discharge observations exist but are insufficient to support local model training or calibration. In these settings, TFT provides a highly effective and scalable forecasting solution, requiring no fine-tuning while delivering accurate and interpretable predictions. Moreover, our findings show that the presence of missing values in discharge inputs does not significantly degrade forecast accuracy, further reinforcing TFT's robustness in data-sparse environments. In summary, the model can be effectively trained on a limited number of hydrologically diverse catchments, encompassing a range of climatic, topographic, and soil conditions, which equips the model with broader exposure to varied runoff-generation patterns. This diversity enhances its ability to generalize and perform well when deployed across a larger group of catchments, making it a cost-efficient and powerful solution for large-scale hydrological forecasting, particularly in regions where traditional calibration-based physical models are impractical.

Author Contributions

Abdul A. Khan: supervision, project administration, writing – review and editing, validation. **Elnaz Heidari:** conceptualization, data curation, formal analysis, methodology, writing – review and editing, visualization, investigation, writing – original draft.

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Disclosure

Research Interest Statement: a single transformer model trained on diverse basins can perform well on unseen catchments, delivering accurate, probabilistic flood forecasts and outperforming the National Water Model.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data and code that support the findings of this study are openly available at the following GitHub URL: https://github.com/elnazhdr/research_tft.

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